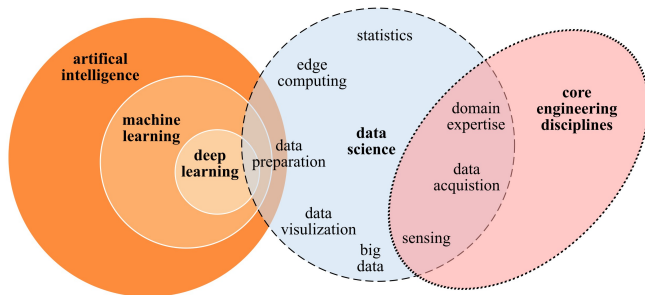


Chapter 1: Basic Concepts in Machine Learning

Austin R.J. Downey

AI, ML, and Data Science



Overlap between the fields of Artificial Intelligence (AI), Data science (DS), and the core engineering disciplines of Civil, Mechanical, Electrical, and Chemical Engineering.

What is Machine Learning?

- ▶ Learns patterns from data
- ▶ Makes predictions or decisions
- ▶ Not robots
- ▶ Not rule-based programming

Example: Spam filter

Early AI: Expert Systems

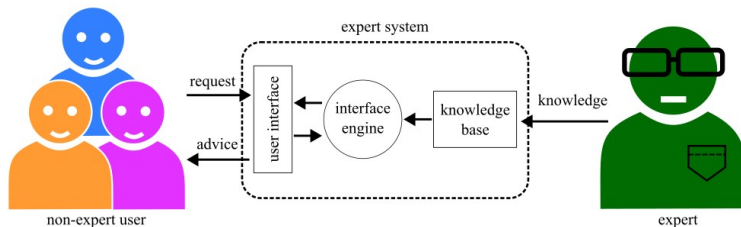


Diagram of an “expert system” in AI, which is a computer program that simulates the decision-making ability of a human expert by using a knowledge base and inference rules.

Examples of Artificial Intelligence

- ▶ **Expert systems:** Simulate decision-making using rules and knowledge bases
- ▶ **Chatbots:** Simulate conversation with human users

Examples of Machine Learning

- ▶ **Regression:** Models relationships between variables
- ▶ **Classification:** Assigns data to categories (e.g., spam vs not spam)
- ▶ **Image/speech recognition:** Identify objects or convert speech to text

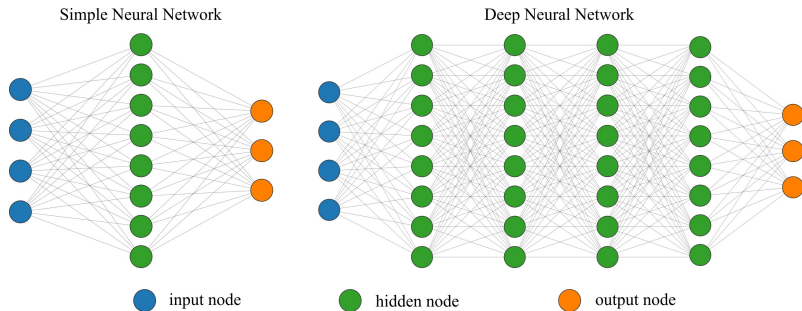
Machine Learning in Practice



A Symbolics Lisp Machine, a specialized hardware platform designed to run expert systems which are a version of AI focusing on answering questions to challenging problems¹.

¹Michael L. Umbricht and Carl R. Friend (Retro-Computing Society of RI), CC BY-SA 3.0
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Deep Learning vs Neural Networks



Simple neural network vs deep learning

The ImageNet Breakthrough (2012)

ImageNet Classification with Deep Convolutional Neural Networks

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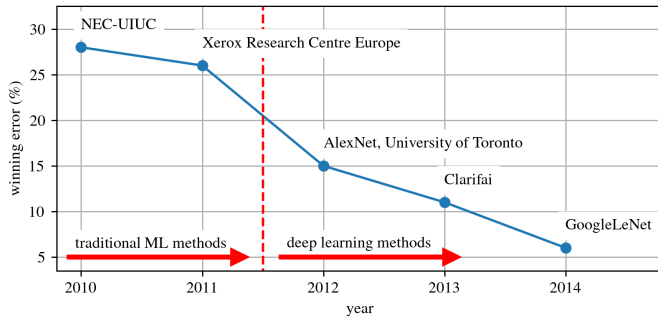
- ▶ AlexNet: 60M parameters, 650k neurons
- ▶ Enabled by GPUs + dropout
- ▶ Top-5 error: 15.3

Abstract

We trained a large, deep convolutional neural network to classify the 1.2 million high-resolution images in the ImageNet LSVRC-2010 contest into the 1000 different classes. On the test data, we achieved top-1 and top-5 error rates of 37.5% and 17.0% which is considerably better than the previous state-of-the-art. The neural network, which has 60 million parameters and 650,000 neurons, consists of five convolutional layers, some of which are followed by max-pooling layers, and three fully-connected layers with a final 1000-way softmax. To make training faster, we used non-saturating neurons and a very efficient GPU implementation of the convolution operation. To reduce overfitting in the fully-connected layers we employed a recently-developed regularization method called “dropout” that proved to be very effective. We also entered a variant of this model in the ILSVRC-2012 competition and achieved a winning top-5 test error rate of 15.3%, compared to 26.2% achieved by the second-best entry.

Geoffrey’s 2012 paper “ImageNet Classification with Deep Convolutional Neural Networks”.

Impact of Deep Learning



ImageNet Competition Results showing the impact of deep learning methods on image classification.

Recognition of Deep Learning



The 2024 Nobel Prize in Physics awarded to John Hopfield and Geoffrey Hinton.

Definition of Machine Learning

- ▶ Systems improve with experience
- ▶ Learn from data, not explicit rules
- ▶ **Arthur Samuel (1959):** “Field of study that gives computers the ability to learn without being explicitly programmed”
- ▶ **Tom Mitchell (1997):** “A computer program is said to learn from experience E with respect to some task T and some performance measure P , if its performance on T , as measured by P , improves with experience E .”

Learning as a Function

$$Y = f(X)$$

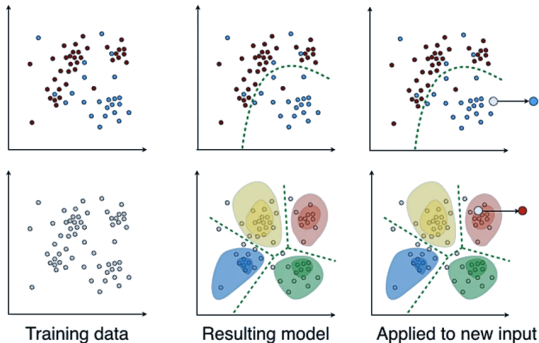
$$Y = f(X) + e$$

- ▶ X : input features
- ▶ Y : output (target)
- ▶ f : unknown function
- ▶ e : error (noise)

- ▶ Training: build the model
- ▶ Inference: use the model

Supervised vs Unsupervised

Supervised learning: each training example has a ground truth label. The model learns a decision boundary and replicates the labeling on new data.



Supervised vs unsupervised machine learning methods¹.

¹ Modified from: Langs, G., Röhrich, S., Hofmanninger, J. et al., CC BY 4.0
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Supervised Learning

Key idea: Data comes with labels

Learn a mapping from inputs $\rightarrow$ outputs

Two main tasks:

- ▶ Classification (categories)
- ▶ Regression (numeric values)

Goal: generalize to new, unseen data

Supervised Learning in Practice

Training uses labeled data, where each example includes input features and a known target.

Examples:

- ▶ Spam filter (spam vs ham)
- ▶ House price prediction

Common algorithms:

- ▶ k-Nearest Neighbors
- ▶ Linear Regression
- ▶ Logistic Regression
- ▶ Support Vector Machines (SVMs)
- ▶ Decision Trees / Random Forests

Unsupervised Learning

Unsupervised learning uses unlabeled data to discover structure and patterns automatically.

Key tasks:

- ▶ **Clustering:** group similar data
- ▶ **Association:** discover relationships
- ▶ **Dimension reduction:** simplify data

Common algorithms:

- ▶ k-Means
- ▶ Hierarchical Clustering
- ▶ Expectation Maximization

Unsupervised Learning Example

- ▶ Finds hidden structure
- ▶ No labels provided
- ▶ Visualization reveals patterns
- ▶ Example: text data
- ▶ Words grouped by frequency/meaning



A semantic word cloud of Barack Obama's First Inaugural Address ¹

¹Modified from: GuoYongzhi, CC BY-SA 4.0 <https://creativecommons.org/licenses/by-sa/4.0>, via Wikimedia Commons

Unsupervised Learning: Concepts and Methods

Unsupervised learning reveals structure in data without labels.

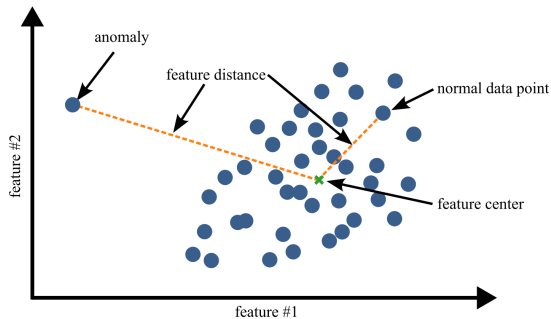
Key concepts:

- ▶ **Clustering:** group similar data points
- ▶ **Association:** discover relationships between variables
- ▶ **Dimension reduction:** reduce number of variables

Common algorithms:

- ▶ k-Means
- ▶ Hierarchical Clustering (HCA)
- ▶ Expectation Maximization

Anomaly Detection



Anomaly Detection for a given set of data.

Detect unusual patterns that differ from normal behavior.

- ▶ Train on normal data
- ▶ Identify outliers or rare events
- ▶ Applications: fraud detection, defects, data cleaning

Semi-Supervised Learning

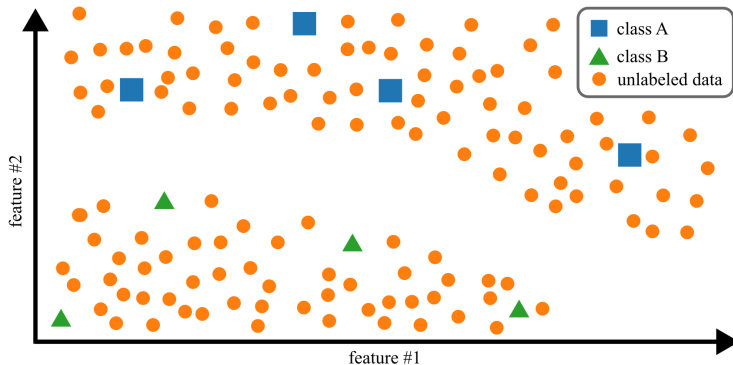
Semi-supervised learning uses a small amount of labeled data with a large amount of unlabeled data.

- ▶ Combines supervised and unsupervised learning
- ▶ Leverages structure in unlabeled data

Example:

- ▶ Group similar images automatically (clustering)
- ▶ Label a few examples
- ▶ Propagate labels to remaining data

Semi-Supervised Learning Example



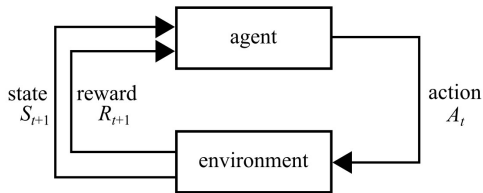
Semisupervised learning enabling the use of a limited set of labeled data to infer the labels of larger unlabeled datasets.

Reinforcement Learning

Reinforcement learning trains an agent to make decisions by interacting with an environment.

- Observe state
- Take action
- Receive reward or penalty

Goal: learn a policy that maximizes total reward

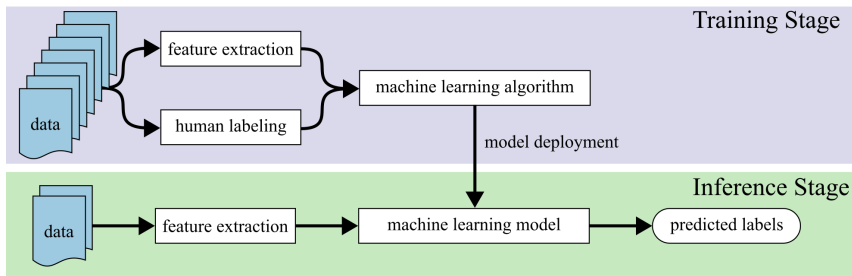


Reinforcement learning diagram.

Types of Learning

- ▶ How models learn from data
- ▶ Two main approaches:
 - ▶ Batch learning (offline)
 - ▶ Online learning (incremental)
- ▶ Trade-off: accuracy vs adaptability vs resources

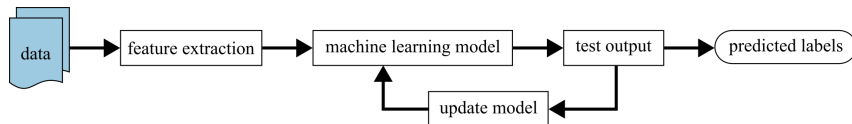
Batch Learning



Batch learning framework with separate training and inference stages.

- ▶ Train once on full dataset
- ▶ No learning after deployment
- ▶ Requires retraining from scratch to update
- ▶ Computationally expensive

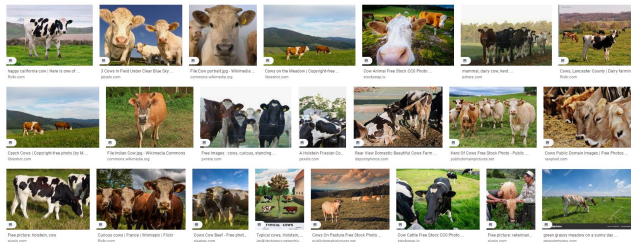
Online Learning



Online learning framework where data is used to continuously training (or update / fine-tune) the model.

- ▶ Learns incrementally from new data
- ▶ Adapts continuously
- ▶ Works for streaming or large datasets
- ▶ Learning rate controls adaptation speed

Bad Data in Machine Learning

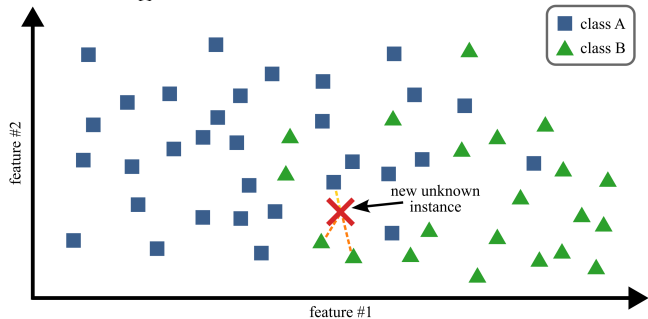


Training a machine learning algorithm to recognize cows may end up just learning to recognize grass. humorously called “short-cut learning”¹.

- ▶ Models learn patterns in data — not concepts
- ▶ Poor or biased data → wrong conclusions
- ▶ Example: learns “grass” instead of “cow”

¹Screen shoot of a Google image search for cows taken by the Austin Downey, all images stated to be under a creative commons license per Google search tools; also assumed to be fair use under given the educational purpose of this text, via <https://www.google.com/>

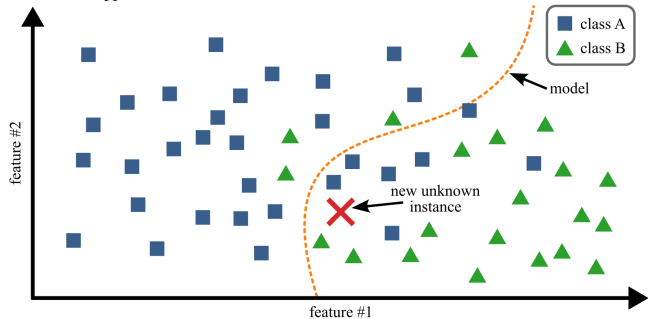
Instance-Based Learning



Instance-based learning where the class of a unknown instance is inferred from the its distance to data points with known labels.

- ▶ Memorizes training examples
- ▶ Uses similarity (distance) to make predictions
- ▶ Flexible but can be slow

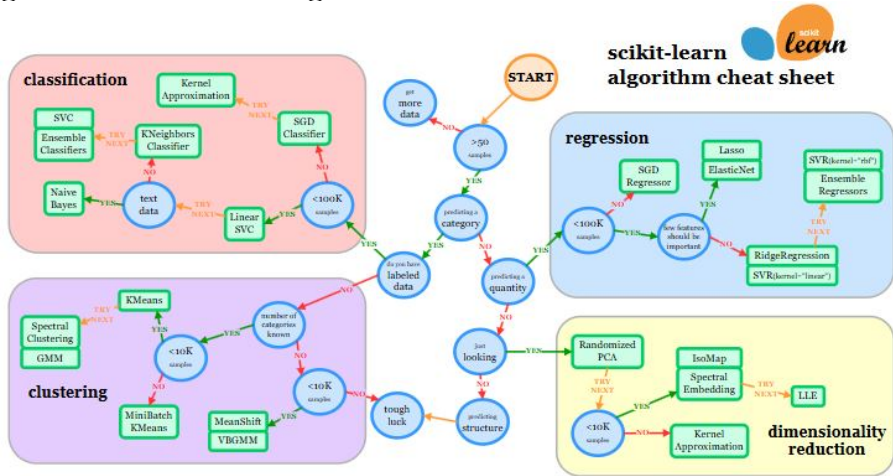
Model-Based Learning



Model-based learning where the class of a unknown instance is inferred from its location in reference to a model trained on the data with known labels.

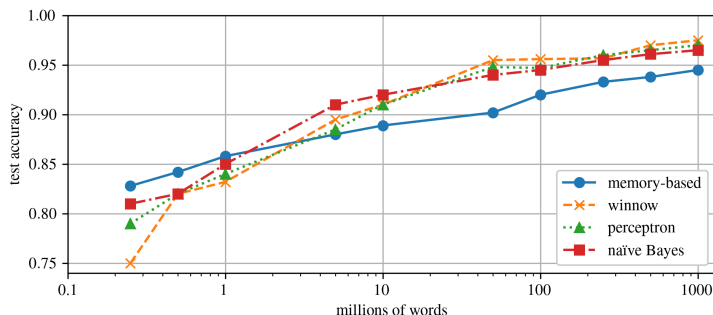
- ▶ Learns a model from data
- ▶ Uses model to make predictions
- ▶ Faster at inference, depends on model quality

Selecting Machine Learning Methods



Flowchart of estimators used in the scikit learn library that intends guide users for what algorithms to use for a given case.

The Unreasonable Effectiveness of Data



Learning curves showing the importance of the amount of data.

- ▶ Michele Banko and Eric Brill (Microsoft, 2001), “Scaling to Very Very Large Corpora for Natural Language Disambiguation”
- ▶ More data $\rightarrow$ better performance
- ▶ Simple models can compete with enough data